Advanced Algebra

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ANSWER KEY — COMPLETE WORKED SOLUTIONS

Part I — Free-Response Questions (FRQs): Foundational Fluency (1–7)

1 **[Foundational]** • **Target: Simplifying rational expressions**

Simplify $\frac{x^2-4}{x^2+x-6}$.

$\frac{x^2-4}{x^2+x-6} = \frac{(x-2)(x+2)}{(x-2)(x+3)} = \frac{x+2}{x+3},\ x \neq 2.$

2 **[Foundational]** • **Target: Simplifying rational expressions**

Simplify $\frac{x^2-5x+6}{x^2-9}$.

$\frac{x^2-5x+6}{x^2-9} = \frac{(x-2)(x-3)}{(x-3)(x+3)} = \frac{x-2}{x+3},\ x \neq 3.$

3 **[Foundational]** • **Target: Simplifying rational expressions**

Combine into a single fraction: $\frac{2}{x+1} + \frac{3}{x-1}$.

$\frac{2}{x+1} + \frac{3}{x-1} = \frac{2(x-1)+3(x+1)}{(x+1)(x-1)} = \frac{2x-2+3x+3}{x^2-1} = \frac{5x+1}{x^2-1}.$

4 **[Foundational]** • **Target: Simplifying rational expressions**

Simplify $\frac{3x^2-12}{x^2-4x+4}$.

$\frac{3x^2-12}{x^2-4x+4} = \frac{3(x-2)(x+2)}{(x-2)^2} = \frac{3(x+2)}{x-2},\ x \neq 2.$

5 **[Foundational]** • **Target: Absolute value equations**

Solve $|2x-5| = 9$.

$2x-5 = 9 \Rightarrow x = 7$. $2x-5 = -9 \Rightarrow x = -2$. Solutions: $x = 7$ or $x = -2$.

6 **[Foundational]** • **Target: Absolute value equations**

Solve $|3x+1| = |x-5|$.

Case 1: $3x+1 = x-5 \Rightarrow 2x = -6 \Rightarrow x = -3$. Case 2: $3x+1 = -(x-5) \Rightarrow 4x = 4 \Rightarrow x = 1$. Both check: solutions $x = -3$ or $x = 1$.

7 **[Foundational]** • **Target: Absolute value equations**

Solve $|x-4| + 3 = 10$.

$|x-4| = 7$, so $x-4 = 7 \Rightarrow x = 11$, or $x-4 = -7 \Rightarrow x = -3$. Solutions: $x = 11$ or $x = -3$.

Part II — Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

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8 [Intermediate] • Target: Absolute value equations

Solve $|4x + 3| = 11$.

$4x + 3 = 11 \Rightarrow x = 2$. $4x + 3 = -11 \Rightarrow x = -3.5$. Solutions: $x = 2$ or $x = -3.5$.

9 [Intermediate] • Target: Polynomial division, remainder & factor theorems

Use the remainder theorem to find the remainder when $f(x) = x^3 + 2x^2 - 5x + 1$ is divided by $(x - 2)$.

$f(2) = 8 + 8 - 10 + 1 = 7$. The remainder is 7.

10 [Intermediate] • Target: Polynomial division, remainder & factor theorems

Given $f(x) = x^3 + 3x^2 - 4x - 12$:

a Use the factor theorem to show that $(x + 3)$ is a factor of $f(x)$.

$f(-3) = (-27) + 3(9) - 4(-3) - 12 = -27 + 27 + 12 - 12 = 0$, so by the factor theorem $(x + 3)$ is a factor.

b Hence factor $f(x)$ completely. Dividing: $f(x) = (x + 3)(x^2 - 4) = (x + 3)(x - 2)(x + 2)$.

(a) $f(-3) = (-27) + 3(9) - 4(-3) - 12 = -27 + 27 + 12 - 12 = 0$, so by the factor theorem $(x + 3)$ is a factor.; (b) Dividing: $f(x) = (x + 3)(x^2 - 4) = (x + 3)(x - 2)(x + 2)$.

11 [Intermediate] • Target: Polynomial division, remainder & factor theorems

Divide $x^3 - 2x^2 - 5x + 6$ by $(x - 3)$ using polynomial long division. State the quotient and remainder.

Quotient $= x^2 + x - 2$, remainder $= 0$. So
$x^3 - 2x^2 - 5x + 6 = (x - 3)(x^2 + x - 2) = (x - 3)(x + 2)(x - 1)$.

12 [Intermediate] • Target: Logarithmic equations

Solve $\log_3(x + 5) = 2$.

$x + 5 = 3^2 = 9 \Rightarrow x = 4$.

13 [Intermediate] • Target: Logarithmic equations

Solve $\log_2(x + 1) + \log_2(x - 1) = 3$.

$\log_2[(x + 1)(x - 1)] = 3 \Rightarrow x^2 - 1 = 8 \Rightarrow x^2 = 9 \Rightarrow x = \pm 3$. Since $x > 1$ is required for $\log_2(x - 1)$ to be defined, reject $x = -3$. Solution: $x = 3$.

14 [Intermediate] • Target: Logarithmic equations

Solve $2\log_5 x = \log_5 16$.

$\log_5(x^2) = \log_5 16 \Rightarrow x^2 = 16 \Rightarrow x = \pm 4$. Since $x > 0$ is required, reject $x = -4$. Solution: $x = 4$.

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Part III — Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

15 [Challenging] • Target: Logarithmic equations • Cross-Lesson: 2.01 Functions: Composite, Inverse & Logarithmic

Solve $\log_3(x-2)=4$.

$x-2=3^4=81\Rightarrow x=83$.

16 [Challenging] • Target: Exponential equations • Cross-Lesson: 3.01 Sequences, Series & Complex Numbers

Solve $3^{2x-1}=27$.

$27=3^3$, so $2x-1=3\Rightarrow x=2$.

17 [Challenging] • Target: Exponential equations • Cross-Lesson: 4.01 Lines, Triangles & Polygons

Solve $5^{x+2}=5^{3x-4}$.

$x+2=3x-4\Rightarrow 6=2x\Rightarrow x=3$.

18 [Challenging] • Target: Exponential equations • Cross-Lesson: 5.01 Trigonometry

Solve $2^x=15$ for x, giving your answer to three decimal places.

$x=\frac{\ln 15}{\ln 2}\approx\frac{2.708}{0.693}\approx 3.907$.

19 [Challenging] • Target: Exponential equations • Cross-Lesson: 6.01 Circles & Coordinate Geometry

Solve $4^x=2^{x+3}$.

Rewrite $4^x=2^{2x}$, so $2x=x+3\Rightarrow x=3$.

20 [Challenging] • Target: Nonlinear systems of equations • Cross-Lesson: 7.01 Solid Geometry & Similarity

Solve the system $y=x^2-1$ and $y=2x+2$.

$x^2-1=2x+2\Rightarrow x^2-2x-3=0\Rightarrow (x-3)(x+1)=0\Rightarrow x=3$ or $x=-1$. Solutions: $(3,8)$ and $(-1,0)$.

Part IV — Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

21 [Foundational] • Target: Simplifying rational expressions

Solve $2^{2x-1}=128$.

a. 5

b. 7

c. 3

d. 4 ✓

$128=2^7$, so $2x-1=7$, giving $x=4$.

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22 [Foundational] • Target: Absolute value equations

Solve $2^{2x-1} = 2$.

a. 0

b. 1 ✓

c. 3

d. 2

$2 = 2^1$, so $2x - 1 = 1$, giving $x = 1$.

23 [Foundational] • Target: Polynomial division, remainder & factor theorems

Solve $2^{2x-1} = 32$.

a. 5

b. 3 ✓

c. 2

d. 4

$32 = 2^5$, so $2x - 1 = 5$, giving $x = 3$.

24 [Foundational] • Target: Logarithmic equations

Solve $3^{2x-1} = 2187$.

a. 3

b. 4 ✓

c. 7

d. 5

$2187 = 3^7$, so $2x - 1 = 7$, giving $x = 4$.

25 [Foundational] • Target: Exponential equations

Solve $7^{2x-1} = 16807$.

a. 4

b. 2

c. 5

d. 3 ✓

$16807 = 7^5$, so $2x - 1 = 5$, giving $x = 3$.

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26 [Foundational] • Target: Nonlinear systems of equations

Solve $5^{2x-1} = 48828125$.

a. 11

b. 5

c. 6 ✓

d. 7

$48828125 = 5^{11}$, so $2x - 1 = 11$, giving $x = 6$.

27 [Foundational] • Target: Rational equations and extraneous solutions

Solve $5^{2x-1} = 125$.

a. 1

b. 3

c. 4

d. 2 ✓

$125 = 5^3$, so $2x - 1 = 3$, giving $x = 2$.

28 [Foundational] • Target: Systems of inequalities and linear programming

Use the remainder theorem to find the remainder when $f(x) = x^3 + 5x^2 - x - 5$ is divided by $(x - 1)$.

a. 0 ✓

b. -5

c. -2

d. 2

$f(1) = 1^3 + 5 - 1 - 5 = 0$. The remainder is 0.

Part V — Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

29 [Intermediate] • Target: A well-designed word problem: production planning at a furniture workshop

Use the remainder theorem to find the remainder when $f(x) = x^3 - x^2 - 2x + 2$ is divided by $(x - (-4))$.

a. -70 ✓

b. -54

c. -72

d. -68

$f(-4) = (-4)^3 - 16 + 8 + 2 = -70$. The remainder is -70.

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30 [Intermediate] • Target: Polynomial Theorems & Synthetic Division

Use the remainder theorem to find the remainder when $f(x) = x^3 + x^2 - 7x + 2$ is divided by $(x - (-4))$.

a. -16

b. -18 ✓

c. -34

d. -20

$f(-4) = (-4)^3 + 16 + 28 + 2 = -18$. The remainder is -18.

31 [Intermediate] • Target: Rational Equations & Partial Fractions

Use the remainder theorem to find the remainder when $f(x) = x^3 - x^2 + 7x - 9$ is divided by $(x - 5)$.

a. 151

b. 128

c. 126 ✓

d. 124

$f(5) = 5^3 - 25 + 35 - 9 = 126$. The remainder is 126.

32 [Intermediate] • Target: Radical Equations & Fractional Exponents

Use the remainder theorem to find the remainder when $f(x) = x^3 - 4x - 4$ is divided by $(x - (-1))$.

a. 1

b. 2

c. -3

d. -1 ✓

$f(-1) = (-1)^3 + 4 - 4 = -1$. The remainder is -1.

33 [Intermediate] • Target: Non-Linear Systems & Inequalities

Use the remainder theorem to find the remainder when $f(x) = x^3 - 3x^2 + 6x + 1$ is divided by $(x - (-1))$.

a. -7

b. -9 ✓

c. -11

d. -6

$f(-1) = (-1)^3 - 3 - 6 + 1 = -9$. The remainder is -9.

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34 [Intermediate] • Target: Simplifying rational expressions

Solve $\log_2(x+3) + \log_2(x-3) = 4$.

a. 7

b. 3

c. -5

d. 5 ✓

$\log_2[(x+3)(x-3)] = 4 \Rightarrow x^2 - 9 = 16 \Rightarrow x^2 = 25 \Rightarrow x = 5$ (rejecting the negative root since $x > 3$ is required).

35 [Intermediate] • Target: Absolute value equations

Solve $\log_2(x+1) + \log_2(x-1) = 3$.

a. -3

b. 5

c. 3 ✓

d. 1

$\log_2[(x+1)(x-1)] = 3 \Rightarrow x^2 - 1 = 8 \Rightarrow x^2 = 9 \Rightarrow x = 3$ (rejecting the negative root since $x > 1$ is required).

36 [Intermediate] • Target: Polynomial division, remainder & factor theorems

Solve $\log_3(x+4) + \log_3(x-4) = 2$.

a. 7

b. -5

c. 3

d. 5 ✓

$\log_3[(x+4)(x-4)] = 2 \Rightarrow x^2 - 16 = 9 \Rightarrow x^2 = 25 \Rightarrow x = 5$ (rejecting the negative root since $x > 4$ is required).

37 [Intermediate] • Target: Logarithmic equations

Solve $5^x = 141$ for x, giving your answer to three decimal places.

a. 2.575

b. 4.075

c. 3.075 ✓

d. 3.575

$x = \frac{\ln 141}{\ln 5} \approx 3.075$.

Part VI — Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

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38 [Challenging] • Target: Exponential equations • Cross-Lesson: 5.01 Trigonometry

Solve $5^x = 410$ for x, giving your answer to three decimal places.

a. 3.738 ✓

b. 4.238

c. 4.738

d. 3.238

$x = \frac{\ln 410}{\ln 5} \approx 3.738$.

39 [Challenging] • Target: Nonlinear systems of equations • Cross-Lesson: 6.01 Circles & Coordinate Geometry

Solve the system $y = x^2 + x + 10$ and $y = -5x + 10$. Find the x-coordinates of the intersection points.

a. $-4, 0$

b. $6, 0$

c. $-6, 0$ ✓

d. $-6, -2$

$x^2 + x + 10 = -5x + 10 \Rightarrow x^2 + 6x = 0 \Rightarrow (x - (-6))(x - 0) = 0 \Rightarrow x = -6$ or $x = 0$.

40 [Challenging] • Target: Rational equations and extraneous solutions • Cross-Lesson: 7.01 Solid Geometry & Similarity

Solve the system $y = x^2 + 13x + 27$ and $y = 4x + 7$. Find the x-coordinates of the intersection points.

a. $-5, -4$ ✓

b. $5, 4$

c. $-3, -4$

d. $-5, -6$

$x^2 + 13x + 27 = 4x + 7 \Rightarrow x^2 + 9x + 20 = 0 \Rightarrow (x - (-5))(x - (-4)) = 0 \Rightarrow x = -5$ or $x = -4$.

41 [Challenging] • Target: Polynomial Theorems & Synthetic Division • Cross-Lesson: 8.01 Statistics & Probability

If $(x - 2)$ is a factor of $P(x) = 2x^3 - 3x^2 + kx - 10$, what is the value of k?

a. 1

b. 2

c. 3 ✓

d. 4

By the Factor Theorem, $P(2) = 0$:

$2(8) - 3(4) + 2k - 10 = 0 \Rightarrow 16 - 12 + 2k - 10 = 0 \Rightarrow 2k - 6 = 0 \Rightarrow k = 3$.

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42 [Challenging] • Target: Rational Equations & Partial Fractions • Cross-Lesson: 2.01 Functions: Composite, Inverse & Logarithmic

Solve the rational equation: $\frac{x}{x-2} + \frac{2}{x+2} = \frac{8}{x^2-4}$.

a. $x = 2$

b. $x = -2$

c. $x = 2$ and $x = -2$

d. **No real solution** ✓

Multiply by $(x-2)(x+2)$: $x(x+2) + 2(x-2) = 8 \Rightarrow x^2 + 4x - 12 = 0 \Rightarrow (x+6)(x-2) = 0$. The roots are $x = -6$ and $x = 2$. However, $x = 2$ is an extraneous root (denominator zero).

43 [Challenging] • Target: Rational Equations & Partial Fractions • Cross-Lesson: 3.01 Sequences, Series & Complex Numbers

Simplify the compound fraction: $\frac{\frac{1}{x+h} - \frac{1}{x}}{h}$ for $h \neq 0$.

a. $-\frac{1}{x(x+h)}$ ✓

b. $\frac{1}{x(x+h)}$

c. $-\frac{1}{x^2}$

d. $\frac{h}{x(x+h)}$

Combine fractions in numerator: $\frac{x-(x+h)}{x(x+h)h} = \frac{-h}{hx(x+h)} = -\frac{1}{x(x+h)}$.

44 [Challenging] • Target: Radical Equations & Fractional Exponents • Cross-Lesson: 4.01 Lines, Triangles & Polygons

If $x^{2/3} - 5x^{1/3} + 6 = 0$, what are all real values of x?

a. 8 **and** 27 ✓

b. 2 and 3

c. ± 8

d. ± 27

Let $u = x^{1/3}$: $u^2 - 5u + 6 = 0 \Rightarrow (u-2)(u-3) = 0 \Rightarrow u = 2$ or $u = 3$. Cubing gives $x = 2^3 = 8$ or $x = 3^3 = 27$.

45 [Challenging] • Target: Non-Linear Systems & Inequalities • Cross-Lesson: 5.01 Trigonometry

Solve the non-linear system: $y = x^2 - 2x - 3$ and $y = 2x + 2$. What is the sum of the x-coordinates of the intersection points?

a. 2

b. 4 ✓

c. -4

d. 6

Set equations equal: $x^2 - 2x - 3 = 2x + 2 \Rightarrow x^2 - 4x - 5 = 0$. By Vieta's formulas, the sum of roots is $-\frac{-4}{1} = 4$.

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